

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

QUIZ 3_Make Up: L7-L11

Question 1

(MRQ) We would like to use polynomial regression to detect if an email is spam or not. The table below shows the training accuracy, validation accuracy and test accuracy when different polynomial orders are used. Which polynomial order should be selected?

order	Training accuracy	Validation accuracy	Test accuracy
1	0.90	0.90	0.98
2	0.94	0.98	0.95
3	0.96	0.98	0.92
4	0.98	0.97	0.96

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (c). Hyperparameter should be chosen according to validation performance. If validation performance is the same, then choose the one with the best training performance.

Question 2

(MCQ) We would like to build a system that predicts if a patient is at risk of high blood pressure. We have a labeled dataset of past records, where each data sample is labeled as “No” or “Yes”. Which type of learning should be used?

- (a) Supervised Learning
- (b) Unsupervised Learning
- (c) Reinforcement Learning
- (d) None

Answer: (a). Labelled data should use supervised learning.

Question 3

Given the following dataset for training:

Sample No.	$\mathbf{x}$	y
1	[1.5, 2.0]	4.3
2	[-0.5, 1.8]	1.2
3	[2.1, -1.2]	3.7
4	[0, 0.5]	0.8

where $\mathbf{x}$ is a 2-dimensional feature vector and y is the corresponding target value. If we fit a polynomial regression model of order 2, we can find a solution $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$ for the regression coefficients.

- (a) True
- (b) False

Answer: (b). $\mathbf{P}^T \mathbf{P}$ is not invertible.

Question 4

(MRQ) In an educational setting, instructors often want to identify students at risk of failing courses so that they can provide timely support. We would like to develop a binomial logistic regression model that can predict whether a student will pass or fail a course based on historical data of student exam scores labelled with 0 if failed or 1 if passed. After training, a new record is present to the model, and the output is 0.8. Which of the following statements is/are correct?

- (a) The model is 80% confident that this student will pass.
- (b) The model predicts that this student will pass, because the output is not 0.
- (c) The model predicts that this student will pass, because the output is greater than 0.5.
- (d) None

Answer: (a)

(b) and (c) are wrong because no threshold is predefined, so we are not sure about the exact class prediction.

Question 5

(MRQ) Which of the following statements is/are correct?

- (a) K-means clustering is suitable for datasets with spiral-shaped clusters.
- (b) Given the same set of data points, the number of well-separated clusters that can be formed is always more than the number of center-based clusters that can be formed.
- (c) Running K-means clustering multiple times with different random initializations can guarantee reaching the global minimum solution.
- (d) Each data point can belong to multiple clusters for K-means clustering.
- (e) None

Answer: (e)

(a) is wrong because K-means clustering is suitable for spherical clusters.

(b) is wrong because the number of well-separated clusters that can be formed is always no more than the number of center-based clusters that can be formed, given the same set of data points.

(c) is wrong because it cannot guarantee reaching the global minimum solution. It only increases the likelihood.

(d) is wrong because each data point can belong to only one cluster for K-means clustering. K-means clustering belongs to hard clustering which assigns each data point to exact one cluster.

Question 6

In Fuzzy C-Means clustering, the membership degrees of a data point across all clusters always sum to 1.

- (a) True
- (b) False

Answer: (a). Proved by Tutorial 7 Q4.

Question 7

(MRQ) Given the following confusion matrix for a 3-class classification model that predicts the species of an iris flower. Which of the following statements is/are correct? Assume class 1, class 2 and class 3 are Iris setosa, Iris versicolor and Iris virginica, respectively.

	P_1 (Predicted)	P_2 (predicted)	P_3 (predicted)
P_1 (actual)	45	20	0
P_2 (actual)	5	30	0
P_3 (actual)	10	10	60

- (a) There are 20 samples from Iris setosa that are wrongly classified. They are known as False Negatives.
- (b) Each class contains the same number of samples.
- (c) The classification accuracy of the model is 75%.
- (d) The precision of the model is 75%.
- (e) None

Answer: (c)

(a) is wrong because False Negatives only exist for binary classification.

(b) is wrong because class 1 contains 65 samples, class 2 contains 35 samples, and class 3 contains 80 samples. The number is not the same.

(d) is wrong because precision only exist for binary classification.

Question 8

Given the following dataset for training:

Sample No.	$\mathbf{x}$	y
1	[6, 14, 55]	7.0
2	[3, 18, 72]	6.5
3	[7, 11, 65]	7.5
4	[2, 16, 58]	4.5

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target value. Fit a polynomial regression model of order 2. What is predicted output of a new data [4, 19, 80]? If the solution $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$ does not exist, use L2 regularization with the regularization parameter $\lambda = 0.1$.

Your Answer____(up to 4 decimal places)

Answer: 8.2522 ($\mathbf{P}^T \mathbf{P}$ is not invertible)

Range: Min: 8.2520 Max: 8.2530

Please see “Quiz 3_makeup_A.ipynb”

Question 9

Given the following dataset for training a logistic regression model for 3-class classification:

Sample No.	$\mathbf{x}$	y
1	[1.2, -0.4, 0.8]	3
2	[-0.6, 2.0, -0.5]	1
3	[0.3, -1.2, 1.7]	2
4	[2.1, 0.5, -0.8]	3

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target class label.

Assume the initial regression coefficient matrix $\mathbf{W} = \begin{bmatrix} w_{0,1} & w_{0,2} & w_{0,3} \\ w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0.02 & -0.01 & 0.03 \\ -0.05 & 0.04 & 0.01 \\ 0.03 & 0.02 & -0.02 \end{bmatrix}$. Gradient descent is used for learning and learning rate is 0.1.

What is the value of the regression coefficient $w_{0,1}$ after the 1st iteration of gradient descent?

Your Answer: ____ (up to 4 decimal places)

Answer: -0.0084

Range: Min: -0.0084 Max: -0.0083

Please see “Quiz 3_makeup_A.ipynb”

Question 10

Given the following dataset for training a multi-layer perceptron (MLP) of 2 layers for 3-class classification:

Sample No.	$\mathbf{x}$	y
1	[1.2, -0.4, 0.8]	3
2	[-0.6, 2.0, -0.5]	1
3	[0.3, -1.2, 1.7]	2
4	[2.1, 0.5, -0.8]	3

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target class label.

The MLP has been constructed as $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \text{Softmax}([\mathbf{1}, \text{ReLU}(\mathbf{x}^T \mathbf{W}^{(1)})] \mathbf{W}^{(2)})$. Assume

the initial weight matrices $\mathbf{W}^{(1)} = \mathbf{W}^{(2)} = \begin{bmatrix} w_{0,1} & w_{0,2} & w_{0,3} \\ w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0.02 & -0.01 & 0.03 \\ -0.05 & 0.04 & 0.01 \\ 0.03 & 0.02 & -0.02 \end{bmatrix}$. Gradient descent is used for learning and learning rate is 0.1.

What is the value of $w_{0,1}$ in $\mathbf{W}^{(1)}$ after the 1st iteration of gradient descent?

Your Answer: $\times 10^{-4}$ (up to 4 decimal places)

Answer: -1.6792

Range: Min: -1.68 Max: -1.67

Please see “Quiz 3_makeup_A.ipynb”

Question 11

(MCQ) A healthcare clinic collects data from 5 patients, including their age, height, and weight, and measures their blood pressure. They want to build a machine learning model to automatically classify patients into low-risk, medium-risk, and high-risk categories for high blood pressure. The data collected is shown on the table below. Which technique can be observed from this dataset that has been used during the data preparation?

Patient	Age	Height	Weight	Blood Pressure	Risk
1	25	163	50	120	[1,0,0]
2	30	175	78	135	[0,1,0]
3	40	168	65	140	[0,0,1]
4	22	180	72	110	[1,0,0]
5	28	174	68	125	[0,1,0]

- (a) Imputation
- (b) One-hot encoding
- (c) Normalization
- (d) Feature selection using Pearson's correlation coefficient.

Answer: (b).

Question 12

A multi-layer perceptron (MLP) for image classification has been constructed as:

$$\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \text{Softmax}([\mathbf{1}, \text{ReLU}([\mathbf{1}, \text{ReLU}(\mathbf{x}^T \mathbf{W}^{(1)})] \mathbf{W}^{(2)})] \mathbf{W}^{(3)})$$

where $\mathbf{x}$ is the input vector augmented with 1, $\mathbf{W}^{(1)}$, $\mathbf{W}^{(2)}$ and $\mathbf{W}^{(3)}$ are weight matrices. Which of the following statements is/are correct?

- (a) This MLP has 3 layers of neurons.
- (b) The number of neurons in each layer of this MLP is equal to the number of rows in the weight matrix for that layer.
- (c) The number of image classes is equal to the number of columns in $\mathbf{W}^{(3)}$.
- (d) The dimension of $\mathbf{x}$ is equal to the dimension of the flattened image.
- (e) None

Answer: (a)(c)

(b) is wrong because the number of neurons in each layer of this MLP should be equal to the number of columns in the weight matrix for that layer.

(d) is wrong because the dimension of $\mathbf{x}$ is equal to the dimension of the flattened image + 1 (for the bias)